
Error Structure Determines Correctness Preference in Contradictory Training Data

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Language models trained on contradictory data often prefer correct answers, yet
2 the mechanism behind this preference is poorly understood. Without such under-
3 standing, we cannot predict when this implicit filtering will fail—a question crit-
4 ical for controlling model behavior on noisy real-world corpora. We hypothesize
5 that next-token prediction, as a compression process, favors whichever answer
6 cluster has lower description length; truth benefits only when errors lack inter-
7 nal structure. We train transformers (3.5M–1B parameters) from scratch on con-
8 trolled corpora where the same problem appears with both correct and incorrect
9 solutions, systematically varying the structure of errors. We demonstrate three
10 findings: (a) when errors are random, models develop a correctness preference
11 scaling from 65% to 85% with model size; (b) when errors follow a single coher-
12 ent alternative rule, this preference vanishes entirely ($\sim 45\text{--}51\%$); (c) the transi-
13 tion is sharp—two competing wrong rules suffice to restore correctness preference
14 ($47\% \rightarrow 78\%$). The pattern reproduces on real Wikipedia text with entity substitu-
15 tion (71% vs 46%) and at 1B scale on real web text (77% vs 47%). These results
16 support the hypothesis that, in controlled contradictory corpora, model preference
17 is driven by the relative compressibility of competing answer systems rather than
18 by truth per se.

19 1 Introduction

20 Real-world training corpora are noisy: the same question may receive contradictory answers across
21 different documents. Yet language models trained on such data tend to prefer correct information—
22 and this implicit filtering is increasingly relied upon in high-stakes applications from medical QA to
23 legal reasoning. If we do not understand *why* this filtering works, we cannot predict *when* it will fail.
24 This is not merely a theoretical concern: coordinated misinformation campaigns, systematic biases
25 in training data, and internally consistent pseudoscientific frameworks all present cases where errors
26 may share enough structure to evade whatever mechanism produces truthful preferences.

27 Several explanations have been offered. Scaling improves factual performance [Kadavath et al.,
28 2022]. RLHF steers outputs toward human preferences. Frequency matters: factual accuracy cor-
29 relates with how often correct information appears in training data [Elazar et al., 2022, Joshi et al.,
30 2024, Kandpal et al., 2023]. Recent work has found truth-correlated internal representations [Burns
31 et al., 2023, Marks and Tegmark, 2024, Bürger et al., 2024] and established that calibrated models
32 must hallucinate at a rate bounded by the corpus’s monofact rate [Kalai and Vempala, 2024]. Yet
33 none of these explain why the training objective would favor one answer over another when both
34 appear at equal frequency in the same format.

35 We propose that the answer lies in the structure of errors. Minimizing cross-entropy is equivalent
36 to minimizing code length [Shannon, 1948, Deletang et al., 2024], linking LLM training to the

37 Minimum Description Length principle [Rissanen, 1978, Grünwald, 2007]. A correct rule system
38 compresses into a compact representation; diverse errors must be memorized individually. But a *co-*
39 *herent* false system—internally consistent, just wrong—compresses equally well, and the preference
40 vanishes.

41 We test this hypothesis in a controlled experimental setting. We train transformers (3.5M–1B pa-
42 rameters) from scratch on corpora where each mathematical problem appears with both correct and
43 incorrect solutions, systematically varying the structure of errors. This design isolates error structure
44 as the independent variable while holding frequency, format, and domain constant.

45 Our contribution is a single experimentally validated hypothesis: **in controlled contradictory cor-**
46 **pora, the compression objective favors consistency, not truth.** We show that (a) random errors
47 produce a correctness preference scaling from 65% to 85%; (b) a single coherent alternative rule
48 eliminates this preference entirely; (c) two competing rules restore it, pinpointing the compressibil-
49 ity boundary; (d) the same contrast reproduces on real Wikipedia text; and (e) the effect persists at
50 1B parameters on real web text (Section 4.7). Whether this principle extends to large-scale pretrain-
51 ing remains an open question that we discuss in Section 5.

52 2 Related Work

53 **Prediction as compression.** The link between prediction and compression traces to Shannon
54 [1948] and was formalized by Solomonoff [1964] and Rissanen’s MDL principle [Rissanen, 1978,
55 Grünwald, 2007]. Deletang et al. [2024] showed that LLMs are universal compressors. Huang et al.
56 [2024] discovered a linear correlation ($r \approx -0.93$) between compression quality and benchmark
57 performance. Wan and Mei [2025] proved that LLM training approximates Solomonoff induction.
58 Pan et al. [2025] linked compression to knowledge acquisition and scaling; Chlon et al. [2025]
59 linked compression failures to hallucination. Contemporaneous work by Marty et al. [2026] formal-
60 izes simplicity bias through MDL, and Aksenov et al. [2026] connects compression to mathematical
61 modeling. We build on the MDL framework by experimentally varying the description length of
62 false answer systems.

63 **Internal representations and world models.** Compression can give rise to structured internal
64 representations: Li et al. [2023a] found board representations in an Othello model, Gurnee and
65 Tegmark [2024] discovered linear space-time representations in Llama-2, and several studies found
66 truth-correlated representations [Marks and Tegmark, 2024, Burns et al., 2023, Li et al., 2023b,
67 Azaria and Mitchell, 2023, Ravfogel et al., 2025]. Our work operates at the behavioral level: we
68 identify data-level conditions under which compression produces a preference for correct comple-
69 tions, leaving activation-level analysis for future work.

70 **Truthfulness and data statistics.** Joshi et al. [2024] linked truthfulness to “personas” in pretrain-
71 ing data. Elazar et al. [2022] demonstrated frequency dependence. Kandpal et al. [2023] showed a
72 direct relationship between document count and accuracy. Li et al. [2024] studied LLM preferences
73 under conflicting knowledge and found that formality and surface quality predict which version
74 the model favors—a consistency-driven explanation complementary to ours. A growing body of
75 work studies knowledge conflicts directly: Xie et al. [2024] surveyed approaches to resolving con-
76 flicts between parametric and contextual knowledge, and Longpre et al. [2021] showed that entity-
77 substituted evidence can override parametric memory. Unlike these analyses, we fix frequency,
78 format, and surface quality, varying only the compressibility of the error system itself—isolating
79 error structure as the causal variable rather than studying post-hoc conflict resolution.

80 **Simplicity bias and noisy labels.** Neural networks prefer simple functions [Valle-Perez et al.,
81 2019, Mingard et al., 2021, Goldblum et al., 2024]. The noisy labels literature directly paral-
82 lels our setup: Zhang et al. [2017] showed memorization of random labels; Rolnick et al. [2017]
83 showed robustness to massive label noise. Grokking connects to compression as a memorization-to-
84 generalization transition [Nanda et al., 2023, Liu et al., 2023]. Our experiments extend this line by
85 showing that “structured noise”—coherent errors—is not filtered out. To our knowledge, systematic
86 variation of error compressibility in a denoising setting has not been studied before.

87 3 Methodology

88 **Overview.** We test the hypothesis that next-token prediction favors whichever answer system has
 89 lower description length, regardless of its truth value. To isolate this, we train models from scratch
 90 on controlled corpora where each problem appears with contradictory answers, varying only the
 91 structure of errors. The core experiments compare random errors (high description length) against
 92 coherent errors (low description length) at equal frequency (Section 4.1), then probe the boundary
 93 with multi-rule errors of intermediate compressibility (Section 4.2). Transfer to natural language is
 94 tested via Wikipedia entity substitution (Section 4.3).

95 3.1 Hypothesis and MDL Framing

96 Consider a corpus where each problem appears with a correct answer (theory T_1) and an alternative
 97 (T_2). An MDL learner [Rissanen, 1978, Grünwald, 2007] minimizes $L(M) + L(D|M)$. When
 98 $K(T_2) \gg K(T_1)$ (random errors), the false system’s description length grows with corpus size,
 99 favoring T_1 . When $K(T_2) \approx K(T_1)$ (coherent errors), both are compact and the learner has no
 100 basis to prefer one over the other. With N competing false rules, a random “selector” (which rule
 101 applies to which problem) adds $\log N$ bits per problem, progressively degrading the false cluster’s
 102 compressibility.

103 Throughout, “truth” means correctness of mathematical derivations and factual accuracy of
 104 Wikipedia paragraphs—models compress text, not reality. The compressibility gap is domain-
 105 dependent and smaller in natural language than in formal math (Section 4.3). Additional limitations
 106 are discussed in Section 6.

107 3.2 Models and Training

108 Custom decoder-only transformers with GPT-2-like architecture (pre-norm, GELU activation, causal
 109 mask). Size labels below are internal designations and do not correspond to OpenAI’s GPT-2 check-
 110 points (e.g., our “large” at 86M is smaller than GPT-2 Small at 117M):

Config	Layers	d_{model}	Heads	Parameters
tiny	4	256	4	~3.5M
small	6	384	6	~12M
medium	8	512	8	~26M
large	12	768	12	~86M

Table 1: Model configurations used in all experiments.

111 Optimizer: AdamW (weight_decay=0.01), cosine decay with linear warmup, lr = $3\text{e-}4$, seq_len =
 112 256, batch_size = 32, 5000 steps. All experiments use 4 random seeds for core conditions (2 where
 113 noted). Learning curves confirm behavioral plateau by step 3000–4000 at all sizes (Appendix F).
 114 Denoising experiments use PyTorch; multi-rule and Wikipedia experiments use MLX. All problems
 115 are generated and verified by SymPy; no human annotation is involved. Full generation templates
 116 are provided in the supplementary material.

117 3.3 Corpus Design

118 **Denoising setup (primary).** A generator creates mathematical problems of four types: multi-step
 119 arithmetic, factoring, equation solving, and differentiation, formatted as step-by-step derivations
 120 in English. The key design: **each problem appears with contradictory answers**—modeling the
 121 scenario where the same question receives conflicting responses. The tokenizer is character-level
 122 (vocab ~57); BPE robustness is verified in Section 4.4.

123 **Wikipedia setup (transfer).** To test generalization beyond formal math, we construct corpora
 124 from 20,000 Wikipedia paragraphs. Using spaCy NER, we create two corruption modes: *random*
 125 *substitution* (each entity replaced with a random entity of the same type) and *coherent substitution*
 126 (a consistent global mapping, e.g., every “France” $\rightarrow$ “Japan”, every “Paris” $\rightarrow$ “Tokyo”). Coherent
 127 substitution preserves grammatical structure and cross-entity consistency within each paragraph;
 128 random substitution breaks co-occurrence patterns (e.g., a paragraph may mention “Kumamoto”

Condition	Correct	Incorrect	Ratio	Purpose
Equal random	1	1 random	1:1	Signal extraction from random noise
Equal coherent	1	1 coherent	1:1	Control: coherent errors eliminate bias
2:1 noise	1	2 random	1:2	Noise tolerance at moderate noise
4:1 noise	1	4 random	1:4	Noise tolerance at high noise

Table 2: Denoising corpus conditions. Each corpus contains 5,000 unique problems.

Size	Params	Random Accuracy	Coherent Accuracy	Random Seeds	Coherent Seeds
tiny	3.5M	65.3% $\pm$ 1.3%	43.5% $\pm$ 2.6%	4	4
small	12M	74.6% $\pm$ 1.6%	44.5% $\pm$ 3.0%	4	4
medium	26M	81.1% $\pm$ 1.2%	45.8% $\pm$ 3.4%	4	4
large	86M	85.2% $\pm$ 2.3%	51.0% $\pm$ 0.8%	2	2

Table 3: Denoising paired evaluation: equal random vs equal coherent errors, across model sizes. All $\pm$ values are standard deviations across random seeds. Random accuracy scales with model size; coherent accuracy remains near chance.

in a context about French history). Both modes substitute entities of matching type, so surface-level grammatical anomalies (gender, case agreement) are minimal. Models train on 50/50 mixes; evaluation uses 2,000 held-out paired paragraphs.

3.4 Evaluation

Paired evaluation (primary). For each problem, a shared prompt is generated with two completions (correct and incorrect). NLL is computed on completion tokens only, conditioned on the prompt. The primary metric is **pair accuracy**: the fraction of pairs where the model assigns lower NLL to the correct completion. This is equivalent to the Common Language Effect Size (CLES; McGraw and Wong, 1992). Length-matched evaluation confirms that accuracy is not driven by completion length differences (within 1 pp across all conditions).

Generative evaluation. Greedy decoding on 500 test prompts with automated SymPy verification. This confirms that the discriminative effect extends to generation (Section 4.5; Table 6).

4 Results

4.1 The Central Contrast: Random vs Coherent Errors

When the same problem appears with both a correct and a random wrong answer, the model learns to prefer the correct one—accuracy reaches 85% at 86M parameters (Table 3, Figure 1). When the wrong answer follows a coherent rule system, the effect disappears: accuracy hovers near 50% at all scales.

Random accuracy increases monotonically with scale (65% $\rightarrow$ 75% $\rightarrow$ 81% $\rightarrow$ 85%). Coherent accuracy converges toward 50% from below, consistent with the MDL prediction that equally compressible systems at equal frequency should be indistinguishable. The below-chance coherent accuracy at tiny ($\sim$ 43%) reflects textual simplicity of the false rules (e.g., dropping terms in derivatives); the compressor prefers whichever system is simpler to encode. The multi-rule experiment (Section 4.2) controls for this surface-form effect.

4.2 Multi-Rule Errors: The Sharp Crossover

The denoising experiments establish two poles: one coherent rule yields chance, random errors yield strong bias. We probe the boundary by training on corpora with N alternative wrong rules per task type, where each problem’s rule is chosen at random. Each rule is compact, but the mapping “problem $\rightarrow$ rule” is unpredictable.

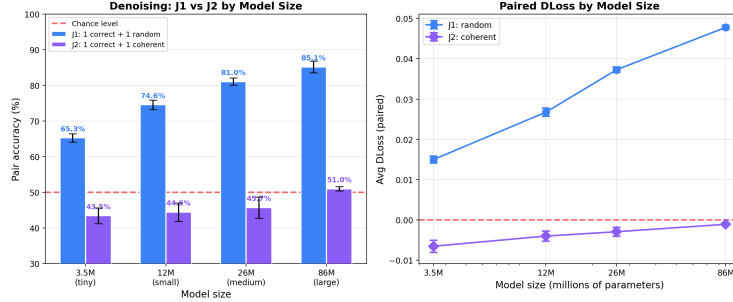


Figure 1: The central contrast. Random errors: accuracy scales from 65% to 85%. Coherent errors: accuracy stays near chance across all sizes.

N Rules	Accuracy	p
1 (coherent)	46.6% $\pm$ 2.4%	~ 1.0
2	77.6% $\pm$ 1.3%	$< 10^{-6}$
3	82.8% $\pm$ 0.1%	$< 10^{-6}$
5	84.8% $\pm$ 0.5%	$< 10^{-6}$
10	88.3% $\pm$ 0.7%	$< 10^{-6}$

Table 4: Multi-rule paired evaluation (tiny, 3.5M, standard 50/50, 4 seeds). The $N=1$ baseline (46.6%) differs from the denoising coherent accuracy in Table 3 (43.5%) because multi-rule experiments use the standard (non-denoising) setup. $N=2$ on small (12M) yields 86.3% $\pm$ 0.8%, confirming the effect strengthens with capacity.

This is the most mechanistically revealing result (Table 4, Figure 2). With one false rule, $K(T_2) \approx K(T_1)$. With two rules, the learner must encode a random selector—which rule applies to which problem—adding high-complexity bits that break compressibility. At $N=10$, accuracy (88.3%) exceeds even the random baseline (79.5%), because multi-rule errors combine compactness of individual rules with incompressibility of the selector.

4.3 Transfer to Real Text: Wikipedia

The random/coherent contrast reproduces (Table 5): random substitution yields 70–71% accuracy (all $p < 10^{-100}$), coherent substitution yields 46–49% (at or below chance). The effect is weaker than in math (71% vs 85%) because natural language provides more textual flexibility for locally fluent errors. Unlike math, Wikipedia accuracy saturates near 70% across the tested range—the compressibility gap in natural language does not substantially benefit from additional capacity at this scale. Per-entity-type accuracy ranges from 82% (LOC) to 61% (CARDINAL), with geographic entities showing the strongest effect (Appendix E).

4.4 Robustness Checks

Tokenization. The primary experiments use a character-level tokenizer (vocab ~ 57). To verify that the effect is not an artifact of tokenization, we repeat the core conditions with a BPE tokenizer (SentencePiece, vocab=1000). The effect survives and strengthens: random accuracy increases from 65.3% to 75.9% in denoising (79.5% to 85.6% in standard); coherent accuracy remains at chance (49.3% and 45.9% respectively). Full results in Appendix F.

Standard (non-denoising) setup. In the standard setup, each problem appears once with either a correct or incorrect solution (no within-problem contradiction). Paired accuracy is higher (79.5–88.3% vs 65.3–85.2%), but the gap closes with scale (from 14 pp at tiny to 3 pp at large). Critically, the random/coherent contrast holds identically: standard coherent accuracy is 47–52%, at chance. Full comparison in Appendix D.

Frequency vs compressibility. For random errors, truth bias persists even when correct examples are a small minority: 67% paired accuracy at 10/90 correct-to-incorrect ratio (Appendix D). For

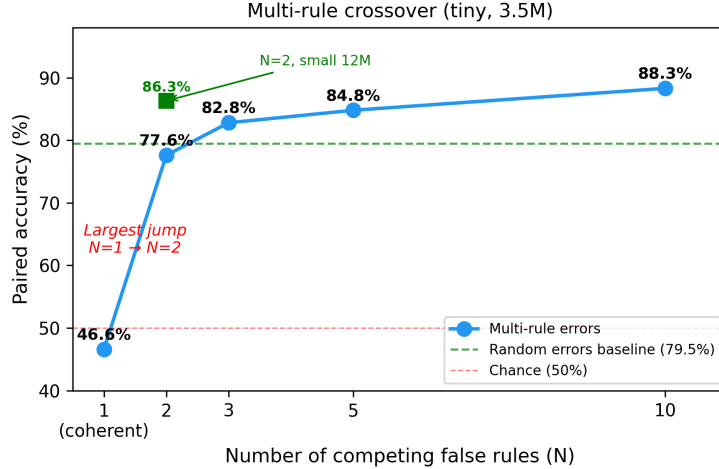


Figure 2: Multi-rule crossover. Accuracy jumps from 47% ($N=1$) to 78% ($N=2$), then grows gradually through $N=10$.

Size	Random Accuracy	Coherent Accuracy
tiny (3.5M)	$69.6\% \pm 0.1\%$	$48.7\% \pm 0.5\%$
small (12M)	$70.7\% \pm 0.4\%$	$46.6\% \pm 0.4\%$
medium (26M)	$71.5\% \pm 0.8\%$	$46.4\% \pm 0.8\%$
large (86M)	$71.4\% \pm 0.7\%$	$45.9\% \pm 1.5\%$

Table 5: Wikipedia entity substitution (50/50, 4 seeds per size). The random/coherent contrast reproduces on real text.

coherent errors, the pattern reverses—frequency alone determines preference: at 40/60 coherent, the model follows the majority with 72% preference for the false system; at 20/80, this rises to 91%. This asymmetry is a direct prediction of the MDL framework: when both systems compress equally well, frequency is the only tiebreaker.

Matched-control ablation. A potential confound is that coherent errors affect more derivation steps per problem (~ 3 on average) than standard random errors (exactly 1). To control for this, we generate “matched random” errors that corrupt the same number of steps as coherent errors but at random positions with random values. Matched-random accuracy is $74.5\% \pm 0.4\%$ (4 seeds)—lower than standard random (79.5%, which corrupts only 1 step) but far above coherent (47.2%, which also corrupts ~ 3 steps). The 27 pp gap between matched-random and coherent at identical step counts confirms that compressibility, not surface-level corruption load, drives the random/coherent contrast.

Noise tolerance. The correct signal persists under increasing noise ratios (1:2 and 1:4 random), with accuracy degrading gracefully from 85% to 66–75% at large scale. Details in Appendix D.

4.5 Generative Evaluation

Paired evaluation is a forced-choice (discriminative) setting. To verify that the effect extends to generation, we run greedy decoding with SymPy verification on 500 problems per model across all sizes.

The gap widens with model size (10 pp at tiny $\rightarrow$ 17 pp at large), confirming that the discriminative preference translates into a generative advantage (Table 6). The absolute generation accuracy is substantially lower than paired accuracy (53% vs 85% at large). This gap reflects the fundamental difference between discrimination and generation: the model may assign higher likelihood to correct completions while lacking the capacity to produce them reliably from scratch. Both metrics agree on the direction: random-trained models outperform coherent-trained ones at all scales.

Size	Random-trained	Coherent-trained	Gap
tiny (3.5M)	30.5% $\pm$ 1.7%	20.8% $\pm$ 3.6%	+9.7 pp
small (12M)	46.7% $\pm$ 1.9%	31.7% $\pm$ 1.8%	+15.0 pp
medium (26M)	50.5% $\pm$ 1.3%	36.1% $\pm$ 2.7%	+14.4 pp
large (86M)	52.8% $\pm$ 1.1%	35.6% $\pm$ 1.8%	+17.2 pp

Table 6: Generative accuracy (4 seeds, 500 problems each). The gap widens with model size.

Condition	seed43	seed44	seed45	Mean
Random	85.3%	85.7%	89.3%	86.8%
Coherent	49.3%	52.8%	49.8%	50.6%

Table 7: Qwen3-0.6B paired evaluation (50/50, 3 seeds). The random/coherent contrast reproduces across architectures.

Per-task breakdown reveals that generation difficulty varies sharply by type. For random-trained large models: derivative (79.5%), algebra (70.4%), equation (61.0%), arithmetic (0%). Multi-step arithmetic requires exact carry propagation across many digits—a well-known difficulty for autoregressive generation at this scale—even though paired evaluation shows strong discrimination. Coherent-trained models show lower generation accuracy across all types, with the gap largest for algebra (70.4% vs 38.3%) and smallest for derivative (79.5% vs 56.5%).

4.6 Architecture Robustness: Qwen3

To verify that the effect is not specific to the GPT-2 architecture, we train a Qwen3-0.6B model (420M parameters, 28 layers, RoPE + GQA + SwiGLU + RMSNorm) from scratch on the same denoising corpora. This architecture differs from GPT-2 in positional encoding, attention mechanism, activation function, and normalization—testing whether the compression-consistency effect is a general property of autoregressive transformers.

The pattern reproduces (Table 7, Figure 3): random errors yield 86.8% accuracy, coherent errors yield 50.6% (chance). The random/coherent contrast is architecture-independent. Qwen3’s random accuracy (86.8%) is comparable to GPT-2 large (85.2%) despite training on the same corpus sized for smaller models, suggesting further gains with compute-matched training.

4.7 Scaling to 1B on Real Text

The preceding experiments use synthetic math corpora and models up to 86M parameters (420M for architecture robustness). A natural objection is that the effect may be specific to toy-scale models or synthetic data. To address both concerns simultaneously, we train a Qwen3-architecture model (~1B parameters) from scratch on 1 GB of FineWeb-Edu [Penedo et al., 2024]—a curated subset of real web text—with mathematical content filtered out and replaced by 8% contradictory math problems (denoising format, equal random or coherent). This setup tests whether the compression-consistency effect survives in a regime where (a) the model has two orders of magnitude more parameters than our primary experiments, and (b) the contradictory signal is a small fraction of a large, naturalistic corpus.

The result (Table 8) confirms that the random/coherent contrast holds: random errors yield 76.8% paired accuracy ($\Delta\text{Loss} = +0.098$, $p < 10^{-6}$), while coherent errors yield 46.7% (chance). The random accuracy is lower than the math-only GPT-2 large (85.2%) and Qwen3-0.6B (86.8%), likely because the contradictory math signal constitutes only 8% of the training corpus, diluted by general text. The key finding is that even in this diluted, naturalistic setting, the compression objective still distinguishes random from coherent errors—and the ΔLoss magnitude (+0.098) is the largest we observe, suggesting that the 1B model develops a stronger per-pair preference despite lower aggregate accuracy.

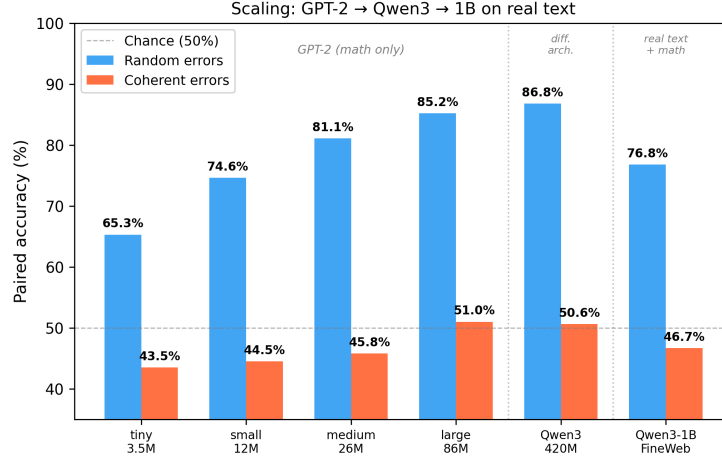


Figure 3: Full scaling results. GPT-2 family (3.5M–86M, math-only), Qwen3-0.6B (420M, different architecture, math-only), and a Qwen3-architecture model (~1B, FineWeb-Edu + 8% math).

Condition	Accuracy	ΔLoss	p	Seeds
Random	76.8%	+0.098	$< 10^{-6}$	1
Coherent	46.7%	−0.003	~ 1.0	1

Table 8: Qwen3-architecture ~1B trained on FineWeb-Edu + 8% contradictory math (denoising, 50/50). The random/coherent contrast reproduces at 1B scale on real text.

5 Discussion

Summary of findings. The experiments support a unified hypothesis: **in our controlled settings, the compression objective tracks consistency rather than truth.** Any internally consistent rule system—true or false—compresses equally well. Truth bias emerges only when false alternatives are structurally incoherent. The evidence converges across setups (denoising, standard, Wikipedia), error types (random, coherent, multi-rule), and evaluation modes (discriminative and generative).

The role of frequency. Our results reveal an asymmetry between random and coherent error regimes. For random errors, compressibility overrides frequency: truth bias persists even at 10/90 correct-to-incorrect ratio (67% accuracy). For coherent errors, frequency is decisive: at 40/60 coherent mixing, the model follows the majority with 72% preference for the false system, rising to 91% at 20/80. This is a direct MDL prediction—when competing systems compress equally well, the more frequent one wins. The practical implication is that coherent falsehood needs no frequency advantage to neutralize truth bias; equal frequency suffices.

What the multi-rule experiment shows. The sharp $N=1 \rightarrow 2$ transition is the strongest evidence that compressibility, not surface-form complexity, is the operative variable. Each individual rule at $N=2$ remains compact, yet the random assignment of rules to problems introduces incompressible bits. This rules out explanations based on lexical diversity or number of corrupted tokens alone.

Implications and scope. In our controlled settings, the training objective does not provide an inherent “truth compass.” Internally coherent falsehood remains competitive. Embedding cross-domain verification can partially restore truth bias, but the effect exhibits inverse scaling: verification accuracy drops from 71% at 3.5M to 61% at 86M parameters (Appendix F), as larger models absorb coherent within-domain patterns more readily. This inverse scaling is verified across 10 models (4 tiny + 4 small + 2 large) with consistent results. This has potential relevance for alignment (systematic false beliefs may resist filtering and grow harder to verify at scale), hallucinations (coherent misconceptions may persist; cf. Kalai and Vempala, 2024), and data curation (diverse errors are filtered effectively, but coordinated errors may not be). The 1B experiment on real web text (Section 4.7) provides initial evidence that the effect persists beyond toy scale and synthetic data, but all our experiments remain far below the scale of production LLMs. Whether these findings ex-

269 tend to large-scale pretraining on heterogeneous real-world corpora remains an open question. We
270 view our results as identifying a mechanism and its boundary conditions, not as direct claims about
271 production LLMs.

272 Our findings admit an interpretive analogy with Popper’s falsifiability criterion [Popper, 1959]: a
273 false theory that requires ad hoc corrections to accommodate observations has higher description
274 length than a true one. The analogy is limited—the model does not “test” theories—but the structural
275 parallel is suggestive.

276 6 Conclusion

277 Whether the compression mechanism that produces truthful preferences in language models extends
278 to large-scale pretraining remains an open question. In our controlled experiments with transformers
279 up to 86M parameters—and up to ~ 1 B parameters on real web text (Section 4.7)—we find that
280 models prefer the correct answer only when errors are structurally incoherent; a single coherent
281 alternative rule eliminates this preference, and two competing rules restore it. The same pattern
282 reproduces on real Wikipedia text, in generative evaluation, across architectures, and at 1B scale on
283 naturalistic corpora. In the settings we study, correctness preference is a compression artifact whose
284 presence or absence is determined by the description length of the false answer system.

285 Limitations

286 **Model scale.** The primary experiments use models of 3.5M–86M parameters. Section 4.7 ex-
287 tends to a Qwen3-architecture model (~ 1 B parameters) trained on real text, confirming the ran-
288 dom/coherent contrast at larger scale. However, this remains far below the scale of production LLMs
289 (7B–400B+), and the 1B result uses only a single seed per condition. For coherent errors, an MDL
290 argument suggests the result should hold regardless of scale: equally compressible systems at equal
291 frequency offer no basis for preference. Replication at multi-billion scale with compute-matched
292 training remains an important next step.

293 **Domain specificity.** Mathematics provides an unusually crisp correct/incorrect distinction. The
294 effect weakens in natural language (71% vs 85%), where errors can remain locally fluent. Extending
295 to domains with competing real-world knowledge systems (e.g., conflicting news sources, historical
296 revisionism) remains open.

297 **Discriminative vs generative gap.** Paired accuracy (85%) substantially exceeds generative accu-
298 racy (53%) at large scale. This gap matters: real-world LLM behavior depends on generation, not
299 discrimination. Several factors may contribute: generation requires producing correct tokens autore-
300 gressively (errors compound), while paired evaluation only requires relative likelihood ranking. The
301 gap narrows with model size (49 pp at tiny, 32 pp at large), suggesting that scale partially bridges
302 it. Critically, both metrics agree on the direction and on the random/coherent contrast at all tested
303 scales—the discriminative preference is not an evaluation artifact. Still, paired accuracy should be
304 interpreted as a controlled diagnostic of model preference, not as a direct measure of output truthful-
305 ness.

306 **Seed counts.** Core conditions use 4 seeds; some conditions use 2. Sufficient for directional stabil-
307 ity, not for tight confidence intervals.

308 **Causal identification.** While we vary error structure as the independent variable, random and co-
309 herent corruptions may differ in additional ways (number of affected derivation steps, surface-form
310 simplicity). The multi-rule experiment partially addresses this by holding individual rule complexity
311 constant, but a fully matched control (identical corruption load, lexical diversity, and output length)
312 would strengthen the causal claim.

313 **Future directions.** Linear probing for “truth directions” vs “coherence directions” in models
314 trained under our conditions could connect behavioral findings to the internal representation liter-
315 ature. Testing interactions with RLHF and investigating whether the compressibility mechanism
316 interacts with in-context learning are natural extensions.

Ethical Considerations

This work demonstrates that, in controlled settings, internally consistent misinformation may be harder for language models to filter than diverse errors. The finding does not directly demonstrate vulnerability in production models. We believe transparent reporting of conditions under which truth bias fails is more beneficial than concealment, as it can inform defensive measures in data curation and model evaluation.

References

- V. Aksenov, E. Bodnia, M. H. Freedman, and M. Mulligan. Compression is all you need: Modeling mathematics. *arXiv preprint arXiv:2603.20396*, 2026.
- A. Azaria and T. Mitchell. The internal state of an LLM knows when it’s lying. In *Findings of EMNLP*, 2023.
- L. Bürger, F. A. Hamprecht, and B. Nadler. Truth is universal: Robust detection of lies in LLMs. In *NeurIPS*, 2024.
- C. Burns, H. Ye, D. Klein, and J. Steinhardt. Discovering latent knowledge in language models without supervision. In *ICLR*, 2023.
- L. Chlon, A. Karim, M. Chlon, and M. Awada. Order sensitivity and information budgeting: Understanding why language models hallucinate. *arXiv preprint arXiv:2509.11208*, 2025.
- G. Deletang, A. Ruoss, J. Grau-Moya, T. Genewein, L. K. Wenliang, E. Catt, S. Legg, et al. Language modeling is compression. In *ICLR*, 2024.
- Y. Elazar, N. Kassner, S. Ravfogel, A. Ravichander, E. Hovy, H. Schütze, and Y. Goldberg. Measuring causal effects of data statistics on language model’s factual predictions. *arXiv preprint arXiv:2207.14251*, 2022.
- M. Goldblum, M. Finzi, K. Rowan, and A. G. Wilson. The no free lunch theorem, Kolmogorov complexity, and the role of inductive biases in machine learning. In *ICML*, 2024.
- P. D. Grünwald. *The Minimum Description Length Principle*. MIT Press, 2007.
- W. Gurnee and M. Tegmark. Language models represent space and time. In *ICLR*, 2024.
- Y. Huang, J. Zhang, Z. Shan, and J. He. Compression represents intelligence linearly. In *COLM*, 2024.
- N. Joshi, J. Rando, A. Saparov, N. Kim, and H. He. Personas as a way to model truthfulness in language models. In *EMNLP*, 2024.
- S. Kadavath, T. Conerly, A. Askell, T. Henighan, D. Drain, E. Perez, J. Kaplan, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- A. T. Kalai and S. S. Vempala. Calibrated language models must hallucinate. In *STOC*, 2024.
- N. Kandpal, H. Deng, A. Roberts, E. Wallace, and C. Raffel. Large language models struggle to learn long-tail knowledge. In *ICML*, 2023.
- J. Li, Y. Cao, S. Huang, and J. Chen. Formality is favored: Unraveling the learning preferences of large language models on data with conflicting knowledge. In *EMNLP*, 2024.
- K. Li, A. K. Hopkins, D. Bau, F. Viegas, H. Pfister, and M. Wattenberg. Emergent world representations: Exploring a sequence model trained on a synthetic task. In *ICLR*, 2023a.
- K. Li, O. Patel, F. Viegas, H. Pfister, and M. Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *NeurIPS*, 2023b.
- Z. Liu, Z. Zhong, and M. Tegmark. Grokking as compression: A nonlinear complexity perspective. *arXiv preprint arXiv:2310.05918*, 2023.

360 S. Longpre, K. Perisetla, A. Chen, N. Ramesh, C. DuBois, and S. Singh. Entity-based knowledge
361 conflicts in question answering. In *EMNLP*, 2021.

362 S. Marks and M. Tegmark. The geometry of truth: Emergent linear structure in large language
363 model representations of true/false datasets. In *COLM*, 2024.

364 T. Marty, E. Elmoznino, L. Gagnon, T. Kasetty, M. Nishikawa-Toomey, S. Mittal, G. Lajoie, and
365 D. Sridhar. A compression perspective on simplicity bias. *arXiv preprint arXiv:2603.25839*,
366 2026.

367 K. O. McGraw and S. P. Wong. A common language effect size statistic. *Psychological Bulletin*,
368 111(2):361–365, 1992.

369 C. Mingard, G. Valle-Perez, D. Sherrington, and A. A. Louis. Is SGD a Bayesian sampler? well,
370 almost. *JMLR*, 2021.

371 N. Nanda, L. Chan, T. Lieberum, J. Smith, and J. Steinhardt. Progress measures for grokking via
372 mechanistic interpretability. In *ICLR*, 2023.

373 Z. Pan, S. Wang, and J. Li. Understanding LLM behaviors via compression: Data generation,
374 knowledge acquisition and scaling laws. *arXiv preprint arXiv:2504.09597*, 2025.

375 G. Penedo, H. Kydlíček, A. Lozhkov, M. Mitchell, T. Wolf, et al. The FineWeb datasets: Decanting
376 the web for the finest text data at scale. *arXiv preprint arXiv:2406.17557*, 2024.

377 K. Popper. *The Logic of Scientific Discovery*. Hutchinson, 1959.

378 S. Ravfogel, G. Yehudai, T. Linzen, A. Bietti, and J. Bruna. Emergence of linear truth encodings in
379 language models. In *NeurIPS*, 2025.

380 J. Rissanen. Modeling by shortest data description. *Automatica*, 14(5):465–471, 1978.

381 D. Rolnick, A. Veit, S. Belongie, and N. Shavit. Deep learning is robust to massive label noise.
382 *arXiv preprint arXiv:1705.10694*, 2017.

383 C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):
384 379–423, 1948.

385 R. J. Solomonoff. A formal theory of inductive inference. *Information and Control*, 7(1):1–22,
386 1964.

387 G. Valle-Perez, C. Q. Camargo, and A. A. Louis. Deep learning generalizes because the parameter-
388 function map is biased towards simple functions. In *ICLR*, 2019.

389 J. Wan and L. Mei. Large language models as computable approximations to Solomonoff induction.
390 *arXiv preprint arXiv:2505.15784*, 2025.

391 J. Xie, K. Zhang, J. Chen, R. Lou, and Y. Su. Adaptive chameleon or stubborn sloth: Revealing the
392 behavior of large language models in knowledge conflicts. In *ICLR*, 2024.

393 C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals. Understanding deep learning requires
394 rethinking generalization. In *ICLR*, 2017.

395 A Reproducibility

396 All code, data generation scripts, and evaluation scripts are available at [anonymous repository, in-
397 cluded as supplementary material]. Denoising experiments were run on Modal.com T4 GPUs using
398 PyTorch. Standard math and Wikipedia experiments were run on Apple Mac M4 (36GB) using
399 MLX (v0.31.0). Qwen3-0.6B experiments were run on Modal.com A10G GPUs; the ~1B Qwen3-
400 architecture experiments were run on Modal.com A100 GPUs. Over 210 models were trained across
401 all conditions; total compute approximately 150 GPU-hours.

402 **Corpus generation examples.** Each problem is generated programmatically and verified by
403 SymPy. Below is a representative training example (denoising, equal random condition):

404 Problem: Expand and simplify $3 * (2x + 5) - 4x$
 405 Step 1: $3 * (2x + 5) = 6x + 15$
 406 Step 2: $6x + 15 - 4x = 2x + 15$
 407 Answer: $2x + 15$

408 The corresponding random-error version replaces a derivation step with a plausible but incorrect
 409 computation (e.g., $6x + 15 - 4x = 2x + 11$). In the coherent condition, a systematic rule is
 410 applied consistently across all problems of the same type (e.g., for distribution: $a(b + c) = ab + c$
 411 instead of $ab + ac$). The paired test set presents the same prompt with both correct and incorrect
 412 completions for NLL comparison.

413 B Formal MDL Predictions

414 We state the theoretical predictions that motivate our experiments. All three describe the behav-
 415 ior of an idealized two-part MDL learner [Rissanen, 1978, Grünwald, 2007], not formal theorems
 416 about neural network training dynamics. Our experiments (Sections 4.1–4.3) test whether gradient-
 417 descent-trained transformers approximate this behavior.

418 **Setup.** Consider a corpus D containing n problems, each appearing with two answers: one gener-
 419 ated by theory T_1 (correct) and one by theory T_2 (alternative). An ideal MDL learner selects the
 420 theory T^* minimizing total description length $L(T) + L(D|T)$.

421 **Prediction 1** (random errors). *If T_1 is a compact rule system with $L(T_1) = O(1)$ and T_2 assigns*
 422 *independent random answers with $L(T_2|D) = O(n)$, then for sufficiently large n the MDL learner*
 423 *prefers T_1 .*

424 *Proof sketch.* The total code length under T_1 is $L(T_1) + L(D|T_1) = O(1) + O(n \cdot c)$ where c
 425 is the per-problem code length given the rule. Under T_2 , the false answers require $O(n)$ bits of
 426 memorization (each answer is independent). For large n , the $O(n)$ memorization cost dominates
 427 any constant advantage T_2 might have, so $L(T_1) + L(D|T_1) < L(T_2) + L(D|T_2)$.

428 **Prediction 2** (coherent errors). *If both T_1 and T_2 are compact rule systems with $L(T_1) \approx L(T_2) =$*
 429 *$O(1)$, then at equal frequency the MDL learner has no basis to prefer one over the other.*

430 *Proof sketch.* Both theories have comparable model complexity $L(T_1) \approx L(T_2)$. Given either
 431 theory, the data encoding cost is the same: $L(D|T_1) \approx L(D|T_2)$, since both produce deterministic
 432 predictions for their respective answer patterns. The total code lengths are approximately equal
 433 regardless of n .

434 **Prediction 3** (multi-rule errors). *If the false system consists of N compact rules $\{r_1, \dots, r_N\}$ with*
 435 *random assignment to problems, then the false system’s description length grows as $O(n \log N)$,*
 436 *restoring the MDL advantage of T_1 .*

437 *Proof sketch.* Each rule r_i is compact: $L(r_i) = O(1)$. However, encoding which rule applies to
 438 which problem requires a selector $s : \{1, \dots, n\} \rightarrow \{1, \dots, N\}$. When assignments are random
 439 (i.i.d. uniform), the selector has entropy $n \log_2 N$ bits and is incompressible. The total false-system
 440 code length is $L(\{r_i\}) + L(s) + L(D|\{r_i\}, s) = O(N) + O(n \log N) + O(n \cdot c)$. For $N \geq 2$, the
 441 $n \log N$ term makes the false system more expensive than T_1 , which requires no selector.

442 C Compression Measure (gzip)

443 To operationalize the MDL argument, we measure compression ratios (gzip level 9) on concatenated
 444 correct vs incorrect completions from each paired test set.

445 Conditions with larger compression gaps produce stronger truth bias. We present this as supporting
 446 evidence; the primary argument rests on the experimental contrasts in Section 4.

Condition	Correct	Incorrect	Delta	Paired Accuracy
random 50/50	0.1627	0.1639	+0.0012	79.5%
coherent 50/50	0.1656	0.1658	+0.0002	47.2%
contradictory	0.1745	0.1744	-0.0001	49.0%
multirule $N=2$	0.1671	0.1710	+0.0038	77.6%
multirule $N=5$	0.1672	0.1730	+0.0057	84.8%
multirule $N=10$	0.1676	0.1752	+0.0076	88.3%

Table 9: Compression ratio and paired accuracy across conditions. Spearman $\rho = 0.68$, $p = 0.042$.

D Standard (Non-Denoising) Experiments

D.1 Standard Corpus Design

In the standard setup, each problem appears once with either a correct or incorrect solution. The two groups do not share prompts.

D.2 Standard vs Denoising

Size	Standard	Denoising	Gap
tiny	79.5%	65.3%	-14.2 pp
small	86.2%	74.6%	-11.6 pp
medium	87.1%	81.1%	-6.0 pp
large	88.3%	85.2%	-3.1 pp

Table 10: Standard vs denoising accuracy (50/50 random). The gap closes with scale.

The denoising setup is harder (within-problem contradiction vs across-corpus mixing), but the gap closes with scale.

D.3 Frequency Effects

Proportion	Random	Coherent
50/50	80%	47.2%
40/60	79%	27.8%
30/70	75%	14.7%
20/80	69%	9.6%
10/90	67%	-

Table 11: Random vs coherent accuracy across proportions (standard, tiny). Random: truth bias persists at 10/90. Coherent: the model follows pure frequency.

D.4 Noise Tolerance

D.5 NLL Distribution

Random errors produce a right-skewed per-pair NLL difference (mean +0.048, median +0.025, 81.5% positive). Coherent errors produce a symmetric distribution (45.5% positive). This explains why pair accuracy and mean Δ Loss can diverge.

E Additional Transfer Experiments

E.1 Synthetic World

A synthetic world with 50 entities, 4 types, 15 deterministic rules. Random 50/50: $57.7\% \pm 1.7\%$. Coherent 50/50: $46.6\% \pm 1.7\%$. Per-type: minerals best (68.7%), potions near chance (49.1%).

Condition	Ratio	Tiny	Small	Medium	Large
Equal random	1:1	65.3%	74.6%	81.1%	85.2%
2:1 noise	1:2	59.0%	68.6%	73.6%	75.2%
4:1 noise	1:4	56.6%	64.9%	66.6%	65.8%

Table 12: Paired accuracy at increasing noise ratios (denoising). Accuracy degrades gracefully; at 4:1 noise, it plateaus at medium/large.

464 E.2 Wikipedia Per-Entity-Type

Entity Type	N Pairs	Accuracy
LOC	45	82.2%
NORP	197	78.7%
GPE	301	77.1%
PERSON	402	69.9%
ORG	664	65.2%
CARDINAL	172	61.0%

Table 13: Wikipedia accuracy by entity type (random substitution, averaged across sizes).

465 E.3 Cross-Domain Falsification

466 Adding correct cross-domain tasks to a coherent corpus selectively increases derivative accuracy
 467 (35% $\rightarrow$ 56% at 25%). The effect is non-monotonic (drops at 50% due to dilution).

468 F Robustness Checks

469 F.1 BPE Tokenization

Setup	Tokenizer	Random	Coherent
Standard	Char	79.5%	47.2%
Standard	BPE	85.6%	45.9%
Denoising	Char	65.3%	43.5%
Denoising	BPE	75.9%	49.3%

Table 14: BPE vs char-level (tiny, 4 seeds). The effect survives BPE and strengthens for random errors.

470 F.2 Chained Verification

471 Embedding verification dependencies in coherent-error tasks restores truth bias: 70.9% at tiny (vs
 472 43% standard coherent). The effect *decreases* with model size (64% at small, 61% at large)—larger
 473 models absorb coherent errors more readily, making verification less effective.

474 F.3 Learning Curves

475 All sizes reach behavioral plateau by step 3000–4000. The large model achieves 88.8% at step 3000,
 476 stable through 5000.

477 G Per-Seed Details

478 NeurIPS Paper Checklist

479 1. Claims

Size	Seed 1	Seed 2	Seed 3	Seed 4	Mean
tiny	64.2%	64.1%	67.0%	65.8%	65.3%
small	75.1%	76.3%	72.6%	74.2%	74.6%
medium	79.6%	80.9%	82.4%	81.3%	81.1%
large	83.5%	86.8%	–	–	85.2%

Table 15: Denoising equal random, per-seed accuracy.

Size	Seed 1	Seed 2	Seed 3	Seed 4	Mean
tiny	40.2%	44.3%	46.3%	43.0%	43.5%
small	44.8%	45.3%	40.3%	47.4%	44.5%
medium	45.8%	40.9%	47.7%	48.6%	45.8%
large	51.6%	50.4%	–	–	51.0%

Table 16: Denoising equal coherent, per-seed accuracy.

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: All claims are explicitly qualified to our controlled experimental setting. The abstract states the scope (3.5M–1B parameters, controlled corpora) and the introduction notes that extension to large-scale pretraining is an open question.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: A dedicated Limitations section discusses model scale, domain specificity, discriminative vs generative gap, seed counts, and causal identification.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [Yes]

Justification: The MDL predictions (Appendix B) are stated as three numbered propositions with explicit assumptions about an idealized MDL learner. They are clearly marked as predictions about idealized learners, not formal theorems about neural network training dynamics. Proof sketches are provided for each.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: All hyperparameters, model architectures, data generation procedures, evaluation protocols, and random seeds are specified in Section 3 and Appendix A. Full generation templates are included in supplementary material.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Anonymous repository included as supplementary material with full code for data generation, training, and evaluation.

6. Experimental setting/details

515 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
 516 rameters, how they were chosen, type of optimizer) necessary to understand the results?

517 Answer: [Yes]

518 Justification: See Section 3.2 (optimizer, lr, schedule, batch size, steps) and Appendix A
 519 (corpus examples, hardware). Train/test separation is guaranteed by seed: test problems
 520 use a different seed (999) than training.

521 **7. Experiment statistical significance**

522 Question: Does the paper report error bars suitably and correctly defined or other appropri-
 523 ate information about the statistical significance of the experiments?

524 Answer: [Yes]

525 Justification: All $\pm$ values are standard deviations across random seeds (not standard er-
 526 rors). p -values are computed via binomial test. The number of seeds is reported for every
 527 condition. Per-seed breakdowns are provided in Appendix G.

528 **8. Experiments compute resources**

529 Question: For each experiment, does the paper provide sufficient information on the com-
 530 puter resources (type of compute workers, memory, time of execution) needed to reproduce
 531 the experiments?

532 Answer: [Yes]

533 Justification: See Appendix A: approximately 150 GPU-hours total on Modal.com
 534 T4/A10G/A100 GPUs and Apple M4 Pro (36GB). Hardware is specified per experiment
 535 type.

536 **9. Code of ethics**

537 Question: Does the research conducted in the paper conform, in every respect, with the
 538 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

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 541 No human subjects, no private data, no dual-use concerns beyond those discussed in Ethical
 542 Considerations.

543 **10. Broader impacts**

544 Question: Does the paper discuss both potential positive societal impacts and negative
 545 societal impacts of the work performed?

546 Answer: [Yes]

547 Justification: The Ethical Considerations section discusses both the defensive value (in-
 548 forming data curation) and the risk (characterizing conditions under which misinformation
 549 evades filtering).

550 **11. Safeguards**

551 Question: Does the paper describe safeguards that have been put in place for responsible
 552 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
 553 image generators, or scraped datasets)?

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555 Justification: All models are small (≤ 1 B parameters) and trained on synthetic or filtered
 556 web data. They have no capabilities beyond the narrow experimental domain and pose no
 557 misuse risk.

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559 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
 560 the paper, properly credited and are the license and terms of use explicitly mentioned and
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562 Answer: [Yes]

563 Justification: Wikipedia data is used under its CC BY-SA license. FineWeb-Edu [Penedo
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 565 SymPy, spaCy) are open source.

566 **13. New assets**

567 Question: Are new assets introduced in the paper well documented and is the documenta-

568 tion provided alongside the assets?

569 Answer: [Yes]

570 Justification: Code and data generation scripts are documented in the supplementary mate-

571 rial with README and inline comments.

572 **14. Crowdsourcing and research with human subjects**

573 Question: For crowdsourcing experiments and research with human subjects, does the pa-

574 per include the full text of instructions given to participants and screenshots, if applicable,

575 as well as details about compensation (if any)?

576 Answer: [N/A]

577 Justification: No human subjects or crowdsourcing involved. All data is generated pro-

578 grammatically.

579 **15. Institutional review board (IRB) approvals or equivalent for research with human**

580 **subjects**

581 Question: Does the paper describe potential risks incurred by study participants, whether

582 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)

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585 Answer: [N/A]

586 Justification: No human subjects involved.

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